

Kelvin waves can raise West Coast sea levels and increase erosion risks

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An AP report forecasts a series of warm-water **pulses** along the United States' West Coast from October through February. Called Kelvin waves, these broad bulges of water could add to flooding and beach **erosion** when combined with high tides, storms and other ocean waves.

Experts forecast a sea-level increase of six inches to a foot as the **pulses** move north along California toward Oregon and Washington. The report describes a travel speed of about six miles per hour. These are predictions, rather than confirmed outcomes for every coastal community.

Kelvin waves do not resemble breaking surf, and they are not tsunamis. Scientist Dillon Amaya describes their shape as more like a large pancake: much wider than it is deep. He says the water bulge extends 12 to 20 miles **offshore** and about 200 meters down. People cannot see it with the naked eye as a distinct wave, and it does not make boats rise and fall like ordinary surface waves.

The process starts with changes in winds along the **equator**. Normally, trade winds push Pacific water west toward Asia. If those winds weaken or reverse, a bulge of water moves back east toward South America. Once it reaches the coastline, it travels north and south along the Americas.

These waves are a common feature of El Niño, a naturally occurring climate pattern involving warmer ocean waters. Experts say the especially strong El Niño described in the report could **intensify** the effects.

The key concern is the starting height of the sea. If water levels are already six inches above normal, a high tide or storm surge begins from that raised **baseline**. A storm surge is a rise in sea level associated with a storm. Combining these effects can bring water farther onto the coast.

Some communities already face shrinking beaches and rising sea levels. US Geological Survey research oceanographer Sean Vitousek says higher water, larger North Pacific waves expected during El Niño and large spring tides could combine to cause exceptional **erosion**. **Erosion** is the wearing away of material such as beach sand. Spring tides are especially large tides; the term does not refer only to the spring season.

The waves also change conditions below the surface. They increase ocean temperatures and push colder, nutrient-rich water deeper, affecting marine life.

As more **pulses** arrive, the raised water level can strengthen the effects of other coastal processes. The resulting impacts depend on how these conditions combine.

Word Treasure

pulses -- Separate movements or surges that arrive in a series.

offshore -- Away from the shore, out at sea.

equator -- The imaginary line around Earth halfway between the poles.

baseline -- A starting level used to understand later changes.

erosion -- The wearing away of material such as beach sand.

intensify -- Make something stronger.

Background you need

A storm surge is a rise in sea level associated with a storm.

Spring tides are especially large tides; their name does not mean they happen only in spring.

El Niño is a naturally occurring climate pattern involving warmer ocean waters.

Break it down

Opens with the forecast and its possible coastal effects.

Gives the predicted rise, route and travel speed.

Uses Amaya's comparison to explain the waves' broad shape.

Explains formation through changing equatorial winds.

Connects El Niño and a raised sea-level baseline to erosion risks.

Ends with effects below the surface and interacting coastal conditions.

Why it matters

The article explains how broad ocean pulses can raise the starting water level for tides and storms, increasing erosion and flooding risks for coastal communities.

Test what you remember

1. What period does the AP report forecast for the series of Kelvin waves?
 - (A) March through July
 - (B) June through September
 - (C) January through April
 - (D) October through February
2. How fast does the report say the pulses travel?
 - (A) About six miles per hour
 - (B) About two miles per hour
 - (C) About ten miles per hour
 - (D) About twenty miles per hour
3. Why does Dillon Amaya compare a Kelvin wave to a large pancake?
 - (A) It is much colder than its surroundings.
 - (B) It is much faster than ordinary winds.
 - (C) It is much steeper than breaking surf.
 - (D) It is much wider than it is deep.
4. What happens after the water bulge reaches South America's coastline?
 - (A) It travels west across the Pacific.
 - (B) It spreads inland along river valleys.
 - (C) It stays still beneath the coastal boats.
 - (D) It travels north and south along the coasts.
5. What does a raised sea-level baseline mean for a storm surge?
 - (A) The surge starts from a higher water level.
 - (B) The surge changes the direction of trade winds.
 - (C) The surge stops larger waves from reaching beaches.
 - (D) The surge carries colder water toward the surface.
6. Which combination does Sean Vitousek say could cause exceptional erosion?
 - (A) Lower water, weaker winds and smaller tides
 - (B) Higher water, larger waves and large spring tides
 - (C) Warmer rivers, deeper bays and steadier trade winds
 - (D) Colder water, wider beaches and slower currents

Answer key (try the quiz first!): 1: D 2: A 3: D 4: D 5: A 6: B

Questions to think about

1. **Dillon Amaya:** He compares Kelvin waves to a large pancake because they are much wider than they are deep, and says they do not make boats rise and fall like ordinary waves.
2. **Sean Vitousek:** He says higher sea levels, larger North Pacific waves expected during El Niño and large spring tides could combine to cause exceptional erosion.

MY THOUGHTS (at least 20 words):